

Machine Learning: Introduction

Prof. Giuseppe Serra

Università degli Studi di Udine

Machine Learning For Big Data

Machine Learning for Big Data (6 credits)

Course slides and Reference Texts: Teams

Office Hours:

- Tuesday 11:00-12:00

Main Topics

The main topics presented in the course are:

- Linear Regression, Logistic Regression, Support Vector Machines, K-Means, hierarchical clustering, DBSCAN, introduction to Neural Networks and Deep Learning.

Exam

Written test (45 minutes)

- Slides, Notes, Books, Laptops, Cell phones, Tablets or other resources (except for calculator) are not allowed.
- The grade of the written test is valid for the academic year of this course (i.e. February 2026)
- Minimum grade to be admitted to the oral discussion: 18

Oral discussion will be focused on topics presented in class and in laboratory

A student, who does not pass the oral discussion, must repeat again the written test.

About Me

2006 - Laurea Degree of Computer Engineering University of Florence

2006 - Visiting Scholar - Carnegie Mellon University (CMU), Pittsburgh (U.S.A.)

2010 - Ph.D. in Computer Engineering, Multimedia and Telec. (University of Florence)

2011 - Visiting Scholar - TELECOM ParisTech, Paris, France

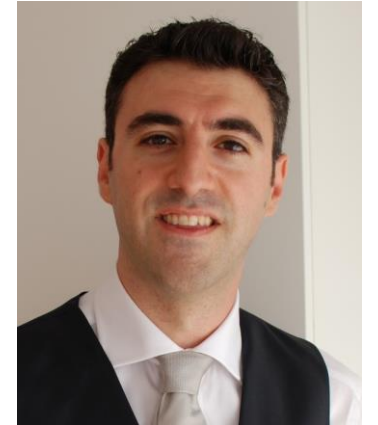
2011 - 2014: Postdoctoral Fellow (University of Florence and Modena and Reggio Emilia)

2014 - 2016 - Assistant Professor of Computer Engineering (University of Modena and Reggio Emilia)

2017 – 2019 - Assistant Professor of Computer Science with tenure track (University of Udine)

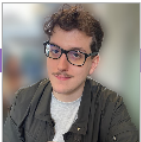
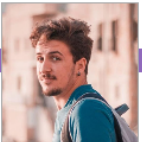
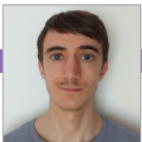
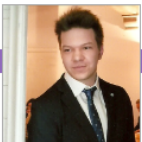
2019 – Associate Professor of Computer Science (University of Udine)

2022 – Co-Founder of the startup Aipertech





<http://ailab.uniud.it/>

 Giuseppe Serra Associate Professor Machine Learning - Deep Learning - Multimedia in en es	 Alex Falcon PostDoc Deep Learning - Vision and Language - Multimedia en in es	 Beatrice Portelli PhD Student Machine Learning - NLP - Adverse Drug Event Extraction en in	 Alessandro Locaputo PhD Student Deep Learning - Digital Humanities - Natural Language Processing en in es	 Mehdi Fasihi PhD Student Deep Learning - Machine Learning en in
 Ali Abdari PhD Student Deep Learning - Vision and Language Understanding - Metaverse in es	 Daniele Lizzio Bosco PhD Student Quantum Machine Learning in es	 Alessandro Tremamunno Intern Deep Learning - NLP en in	 Ian Gallegos Research Fellow Deep Learning - Social Media Understanding - 3D Scene Generation en in	 Franci Rrapi Research Fellow Human Robot Interaction - Artificial Intelligence en in
 Gianluca Macrì PhD Student Deep Learning en in		 Marco Passon Research Fellow Deep Learning - Machine Learning en in		

Machine Learning

Machine Learning is part of our life



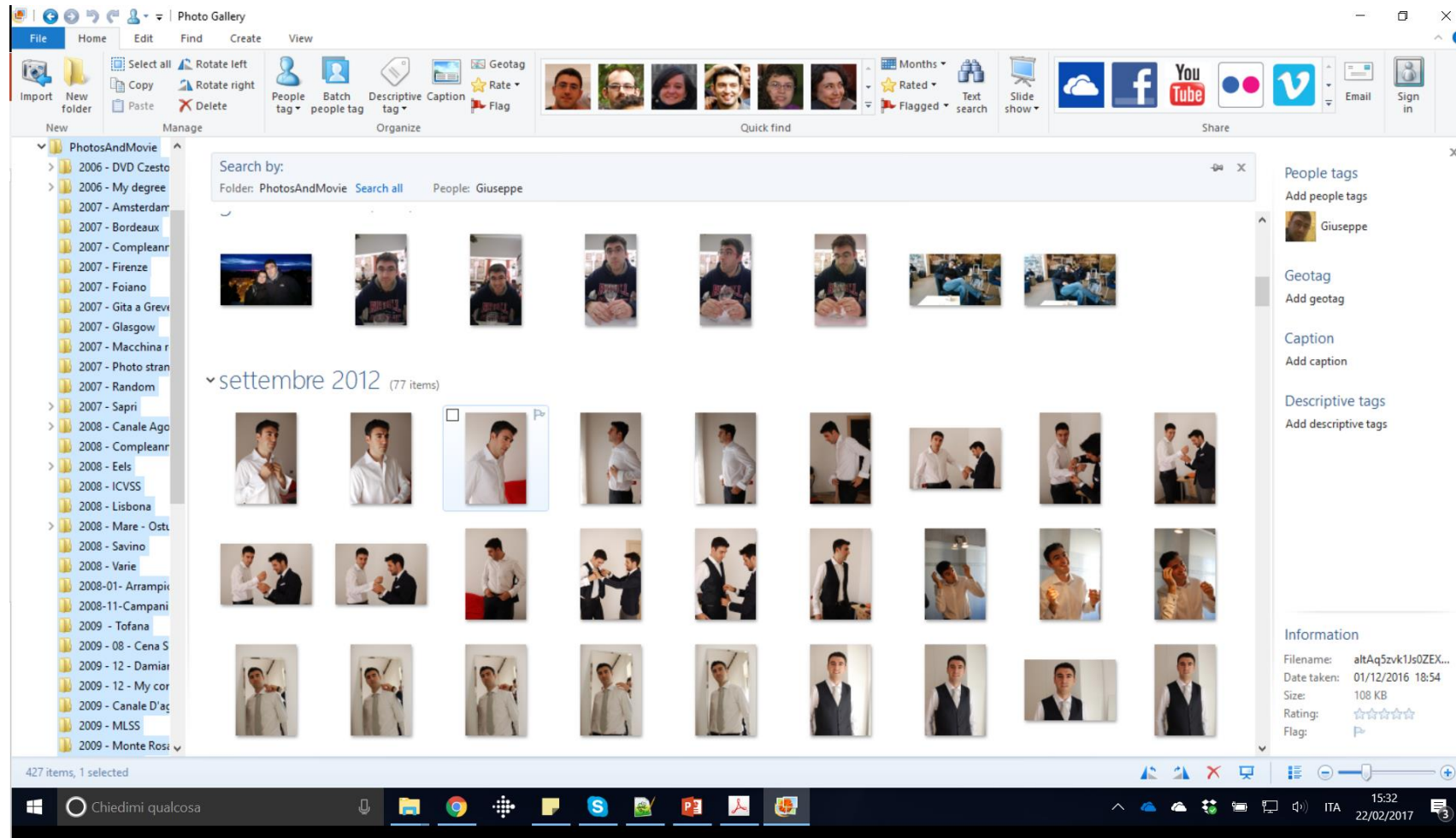
Machine Learning

Machine Learning is part of our life



Machine Learning

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Machine Learning

Applications can't program by hand.
E.g handwriting recognition, Natural Language Processing (NLP), Computer Vision.

Database mining
Large datasets from growth of automation/web.
E.g., Web click data, medical records, biology

Self-customizing programs
E.g., Amazon, Netflix product recommendations

Artificial Intelligence

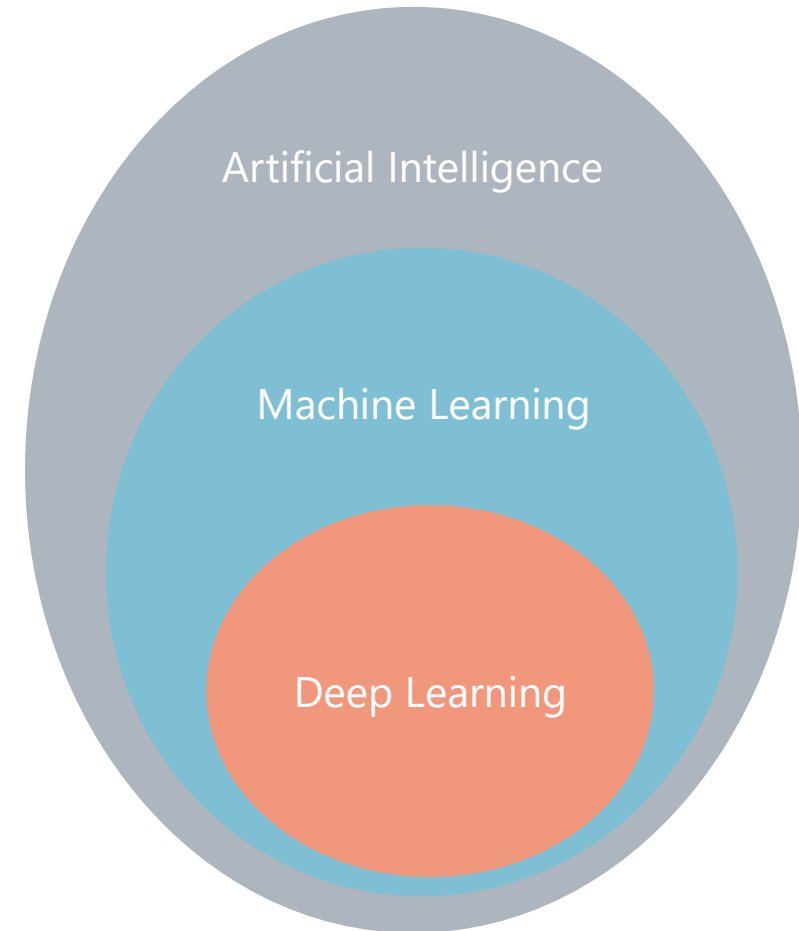
Any technique which enables computers to mimic human behavior.

Machine Learning

Subset of AI techniques which use statistical methods to enable machines to improve with experiences.

Deep Learning

Subset of ML which make the computation of multi-layer neural networks feasible.



What is Machine Learning?

Arthur Samuel (1959). Machine Learning:

- Field of study that gives computers the ability to learn without being explicitly programmed.

Machine learning is a fundamental area of AI and is concerned with the development of systems and algorithms with exploit observations as data

Tom Mitchell (1998) Well-posed Learning Problem:

- A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E .

What is Machine Learning?

Example:

"A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E ."

Suppose your email program watches which emails you do or do not mark as spam, and based on that learns how to better filter spam. What is the task T in this setting? E and P ?

- ☐ Classifying emails as spam or not spam
- ☐ Watching you label emails as spam or not spam
- ☐ The number (or fraction) of emails correctly classified as spam/not spam
- ☐ None of the above—this is not a machine learning problem

What is Machine Learning?

Example:

"A computer program is said to learn from experience E with respect to some task T and some performance measure P , if its performance on T , as measured by P , improves with experience E ."

Suppose your email program watches which emails you do or do not mark as spam, and based on that learns how to better filter spam. What is the task T in this setting? E and P ?

- ☐ Classifying emails as spam or not spam (T)
- ☐ Watching you label emails as spam or not spam (E)
- ☐ The number (or fraction) of emails correctly classified as spam/not spam (P)
- ☐ None of the above—this is not a machine learning problem

Machine Learning

Machine learning algorithms can roughly be divided in:

- Supervised learning
- Unsupervised learning
- Reinforcement learning

Others: recommender systems

Machine Learning Resources

Open source ML Frameworks:

- Scikit-learn
- R
- Weka
- TensorFlow
- PyTorch
- Keras
- Caffe

Research Publications:

- Arxiv

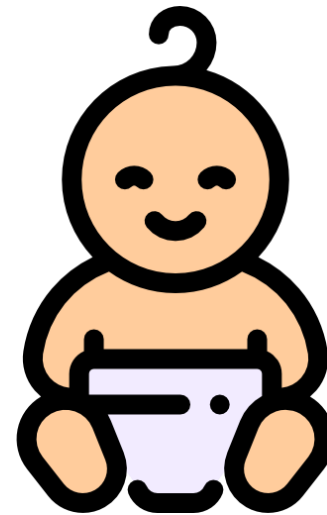
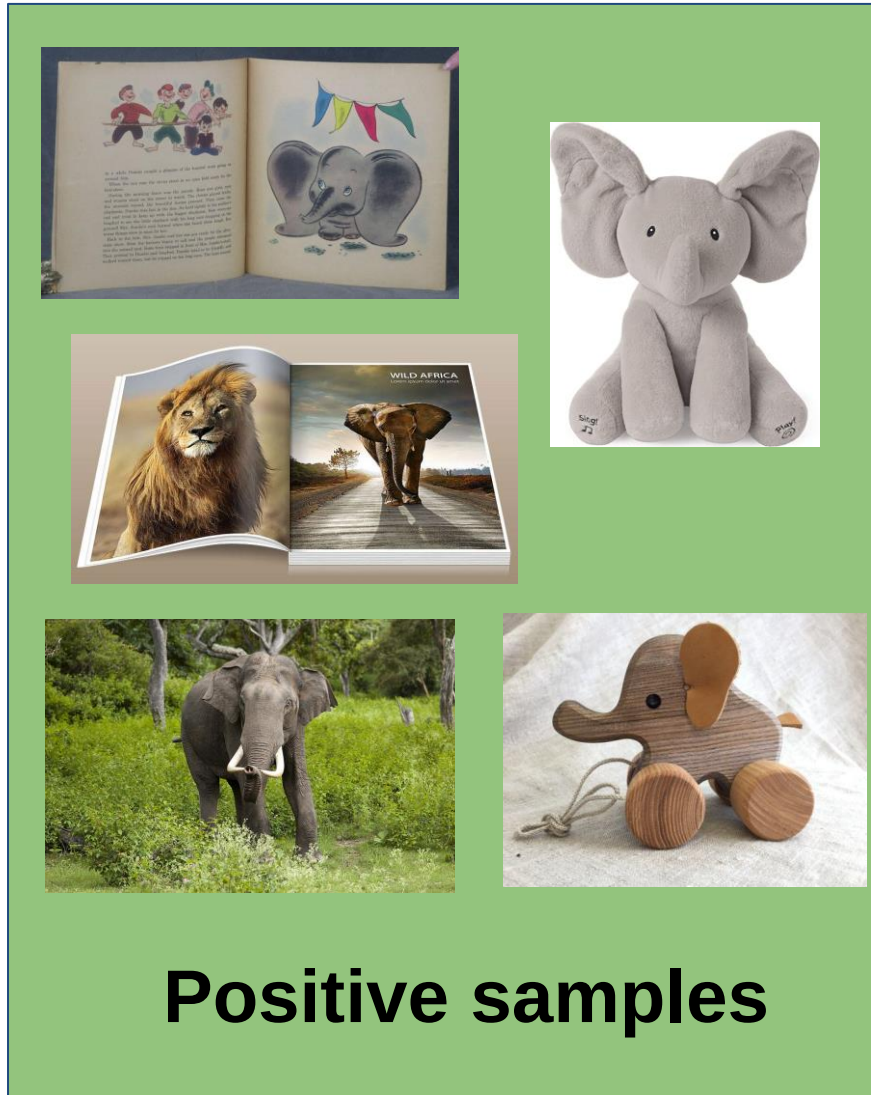
Open source repositories:

- GitHub

Supervised Learning

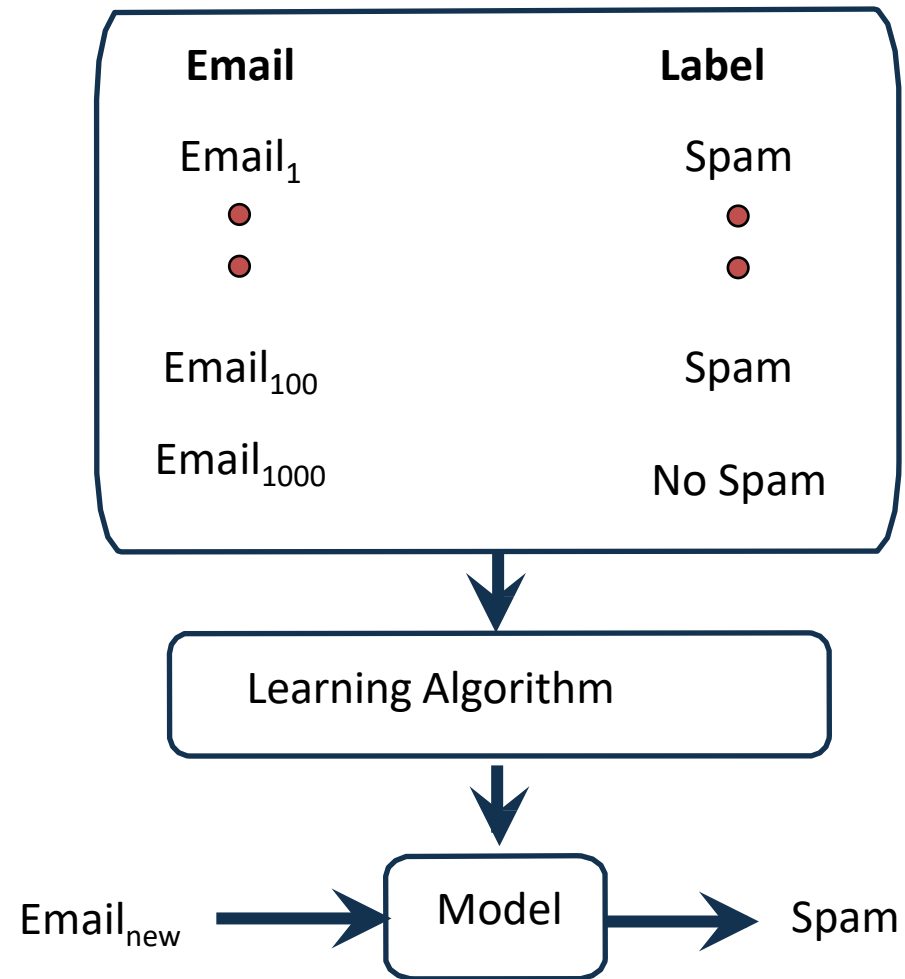
Classification

Teach Computers by samples as a Child does





Spam Detection



Mortgage approval classification

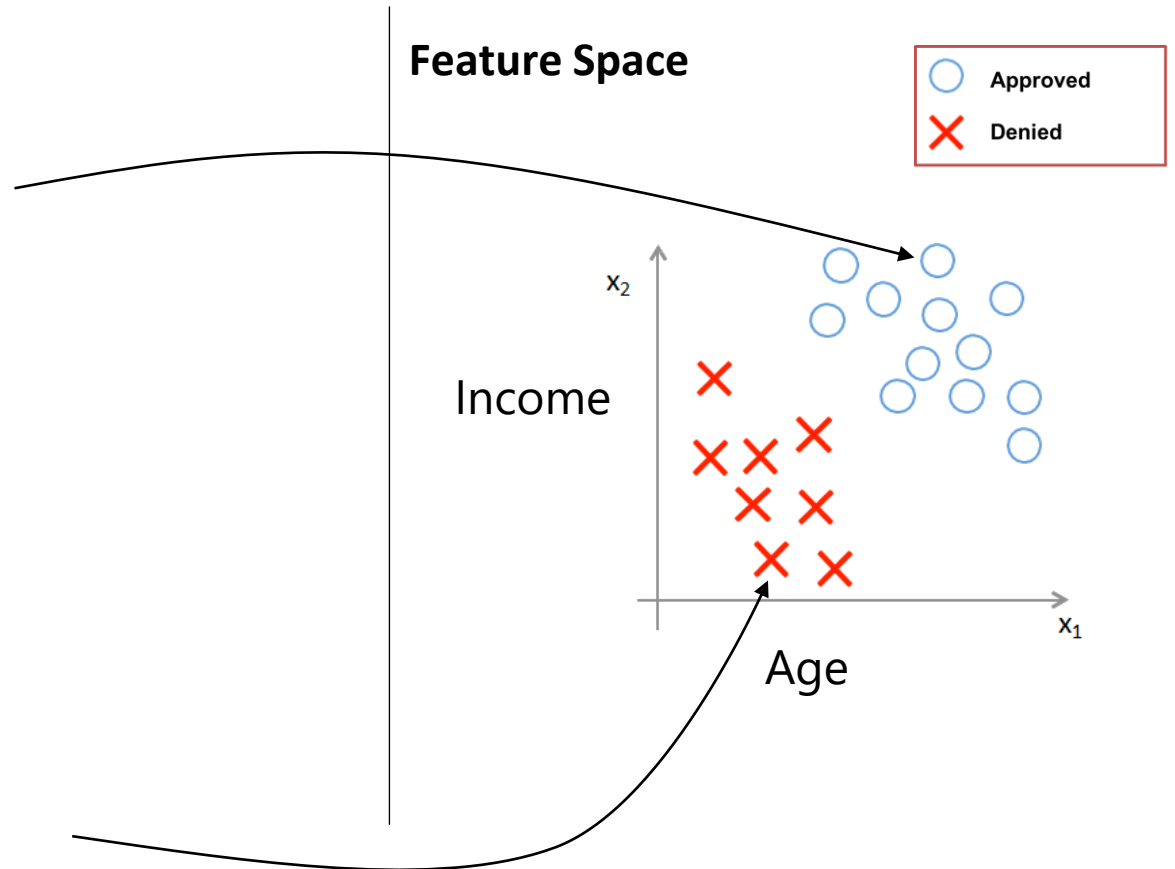
User Feature



Age	Income	Approval
44	36000	yes

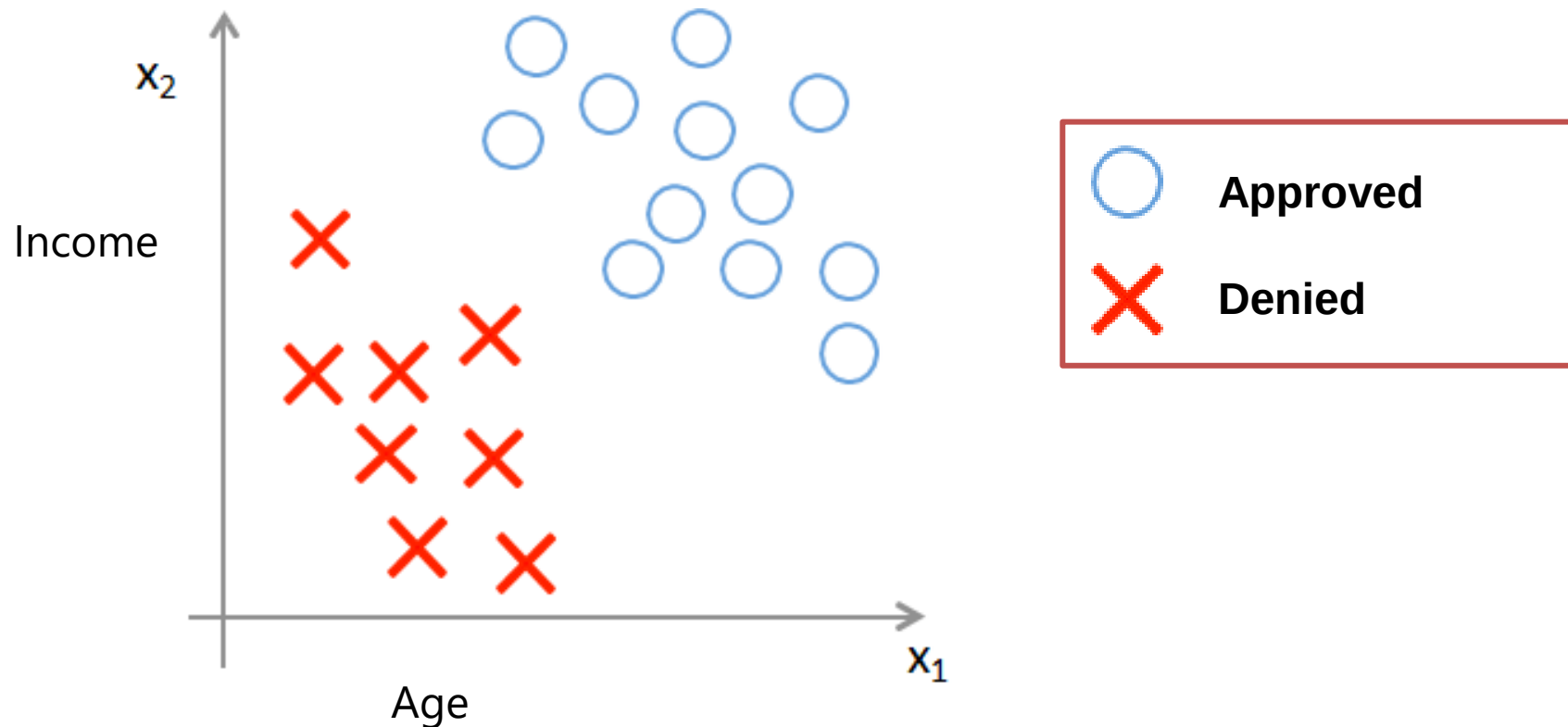


Age	Income	Approval
18	13000	No



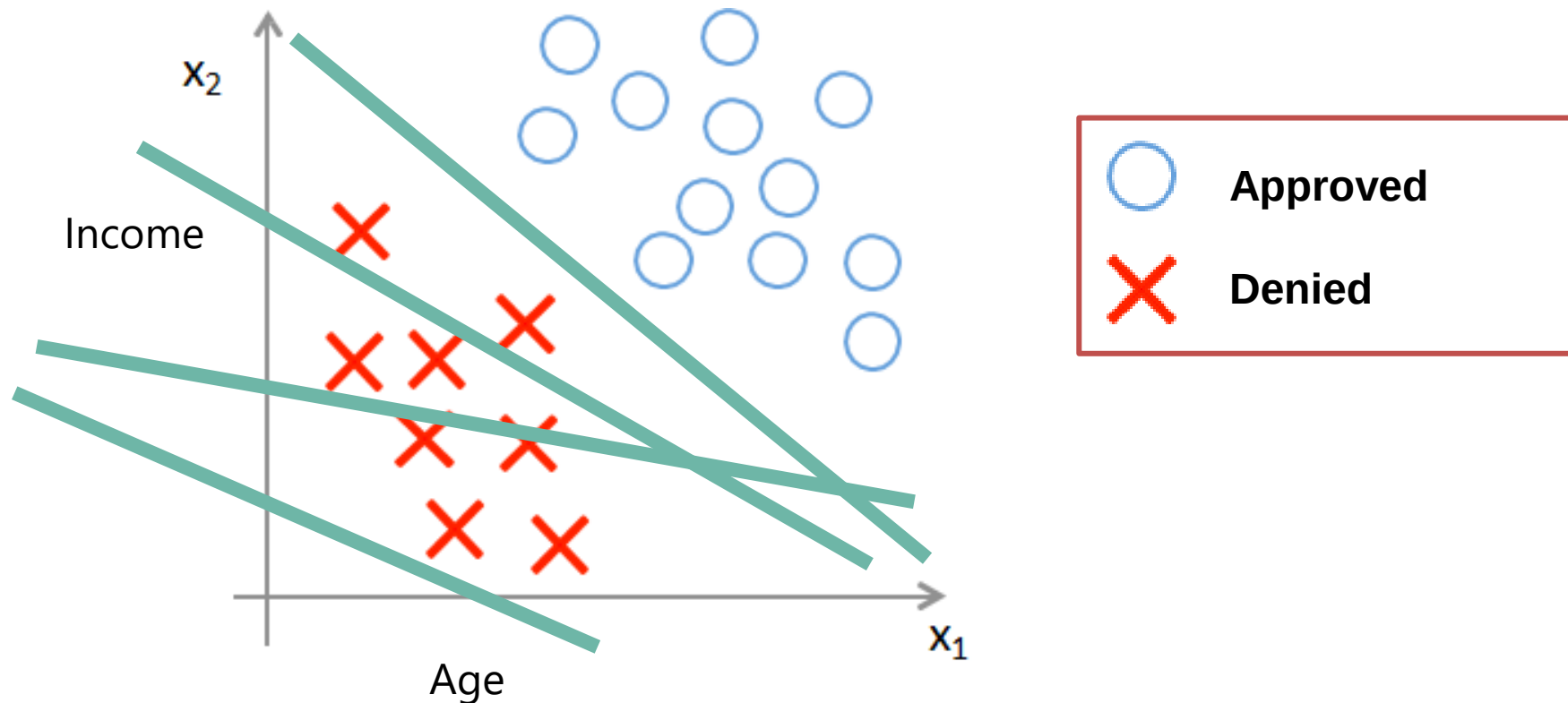
Mortgage approval classification

Suppose you have a dataset of 20 samples:



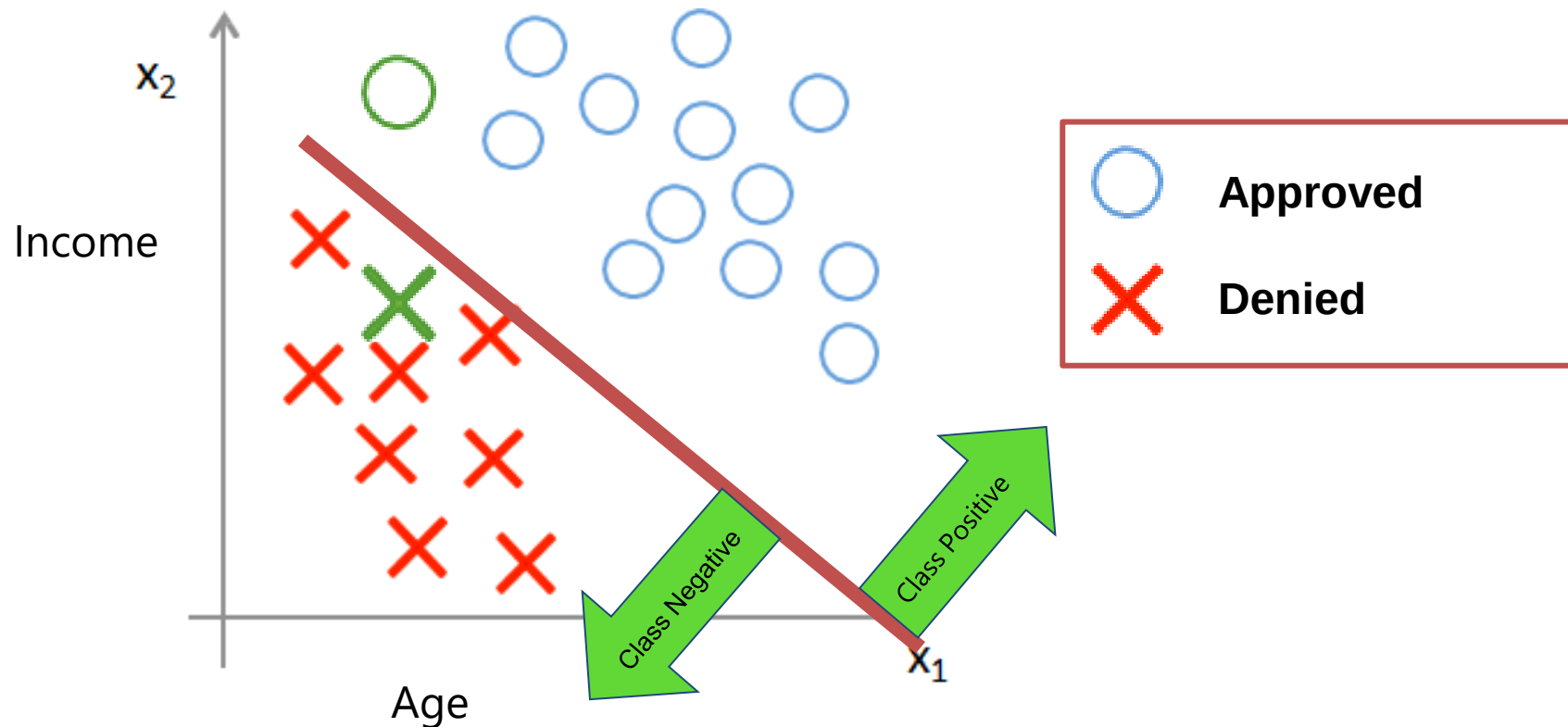
Classificazione dell'approvazione del mutuo

Un algoritmo di classificazione cerca di stimare decision boundary (cioè la linea che separa i dati)

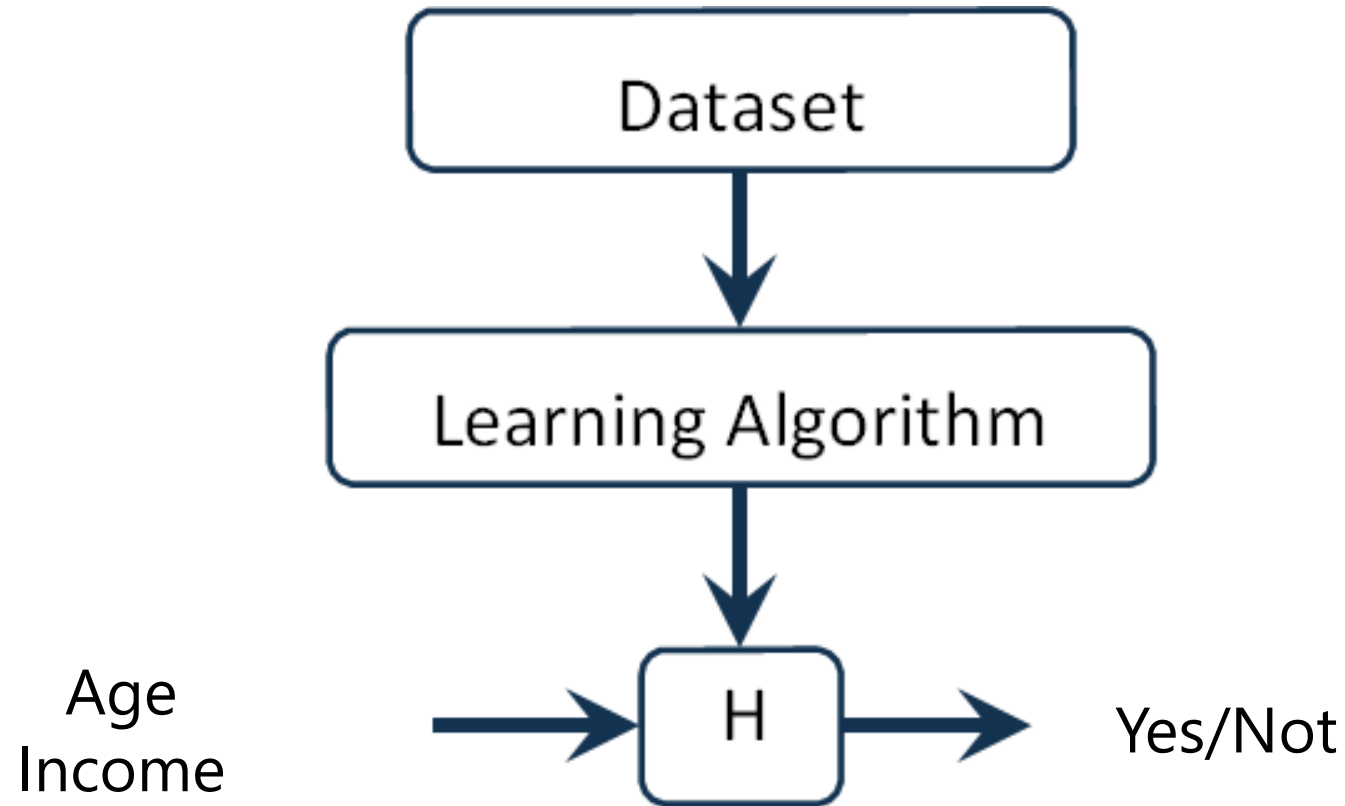


Classificazione dell'approvazione del mutuo

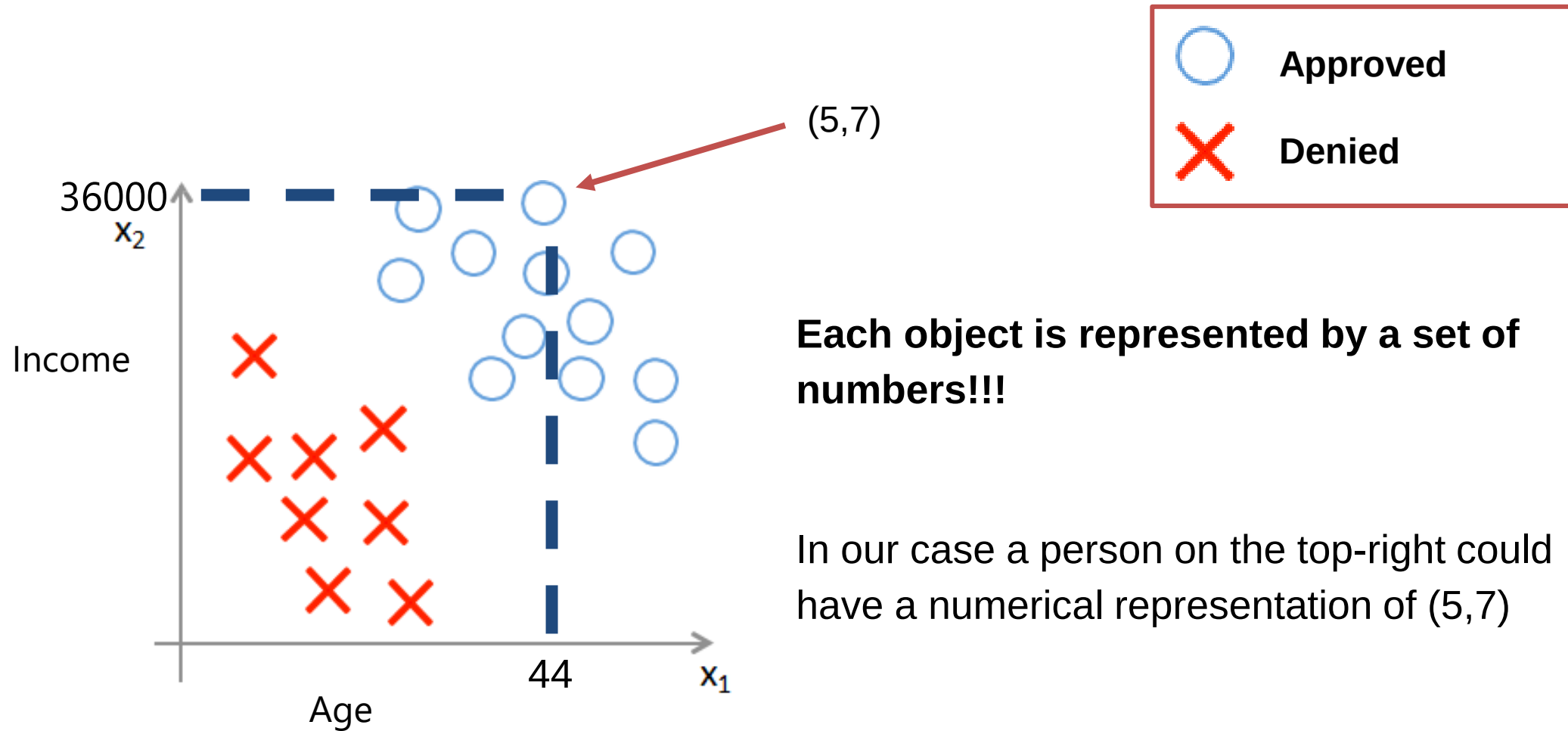
Un algoritmo di classificazione cerca di stimare decision boundary (cioè la linea che separa i dati)



Classificazione dell'approvazione del mutuo



Key Concept #1



Key Concept #2

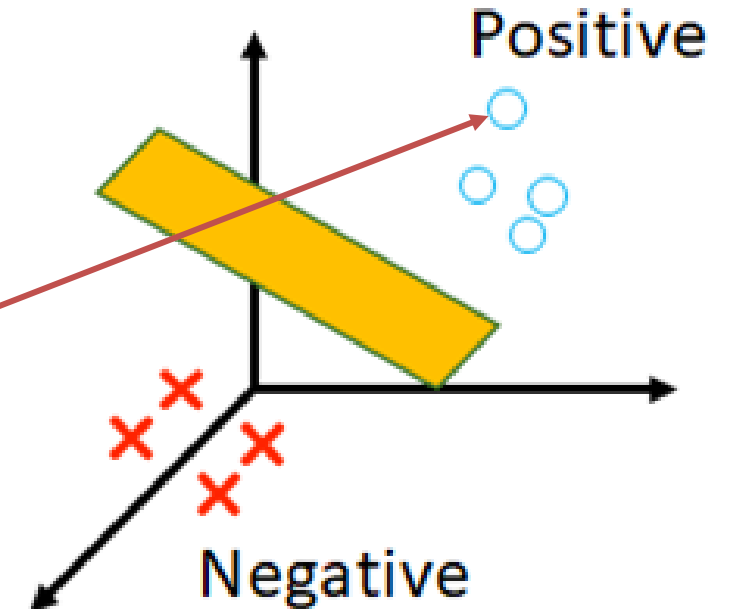
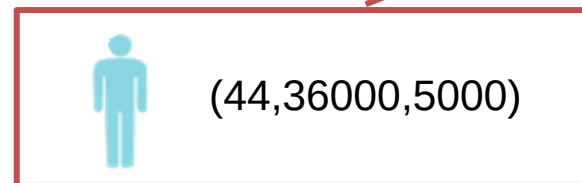
If our objects are represented by 3 features, we need to fit a plane!!!!

If a person is represented with 3 features:

Income;

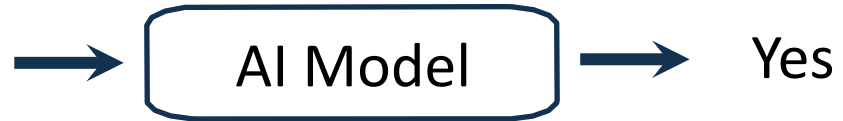
Age;

Current loans



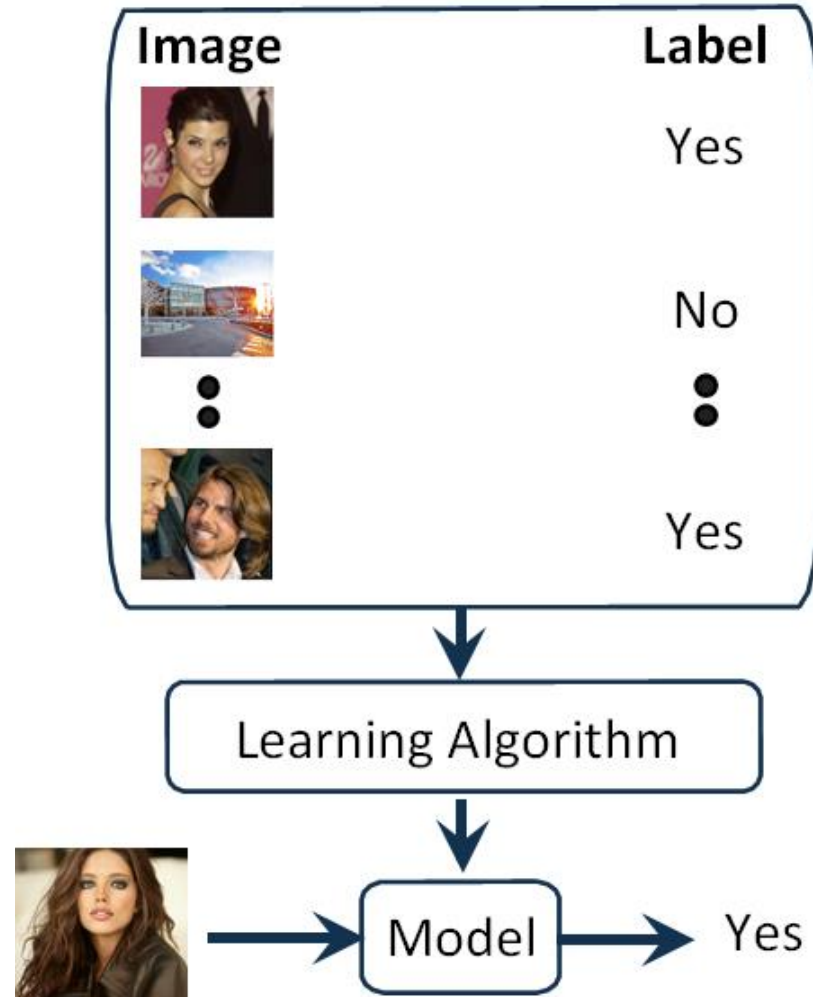
If our objects is represented by multiple features, we need to fit an iperplane!!!!

Face Classification

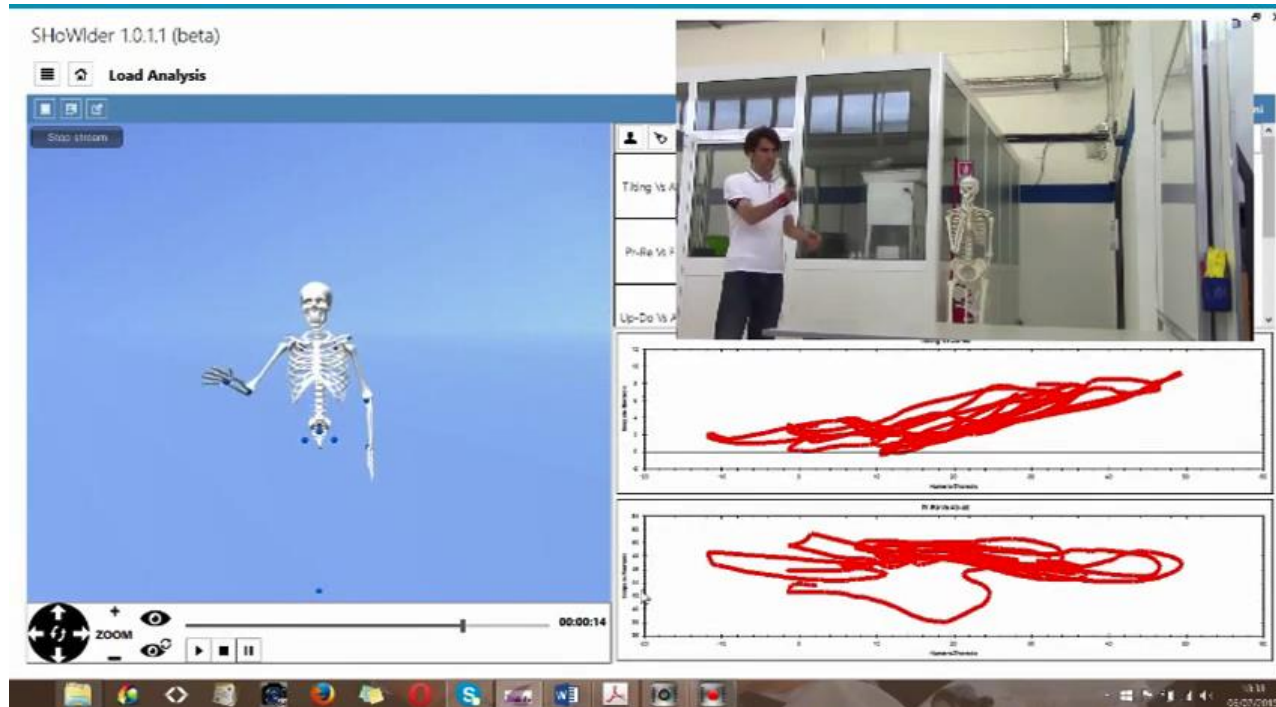


- What data do I need to learn an AI model?

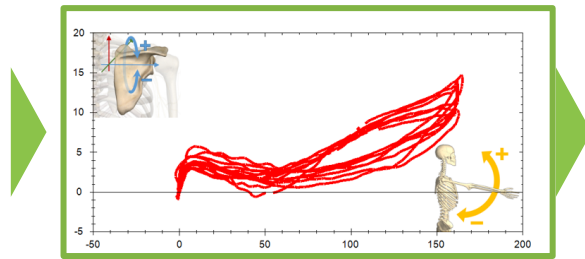
Face Classification



Arm Pathology Analysis



Arm Pathology Analysis



Feature
Extraction

Artificial
Intelligence
Algorithm

"Pathology"

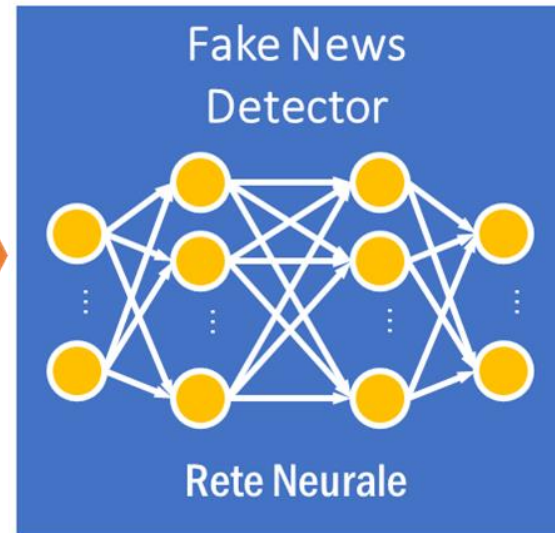
Prediction

"No Pathology"

FAKE NEWS

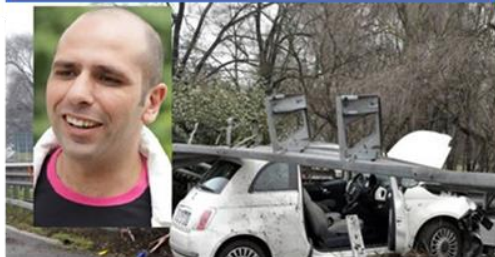
News

In pochi non conoscerranno Checco Zalone. Famoso comico, cabarettista e attore italiano, è stato presente in molte edizioni di Zelig. Proprio poche ore fa è stato coinvolto in un terribile incidente stradale. Le sue condizioni al momento sono critiche [...]



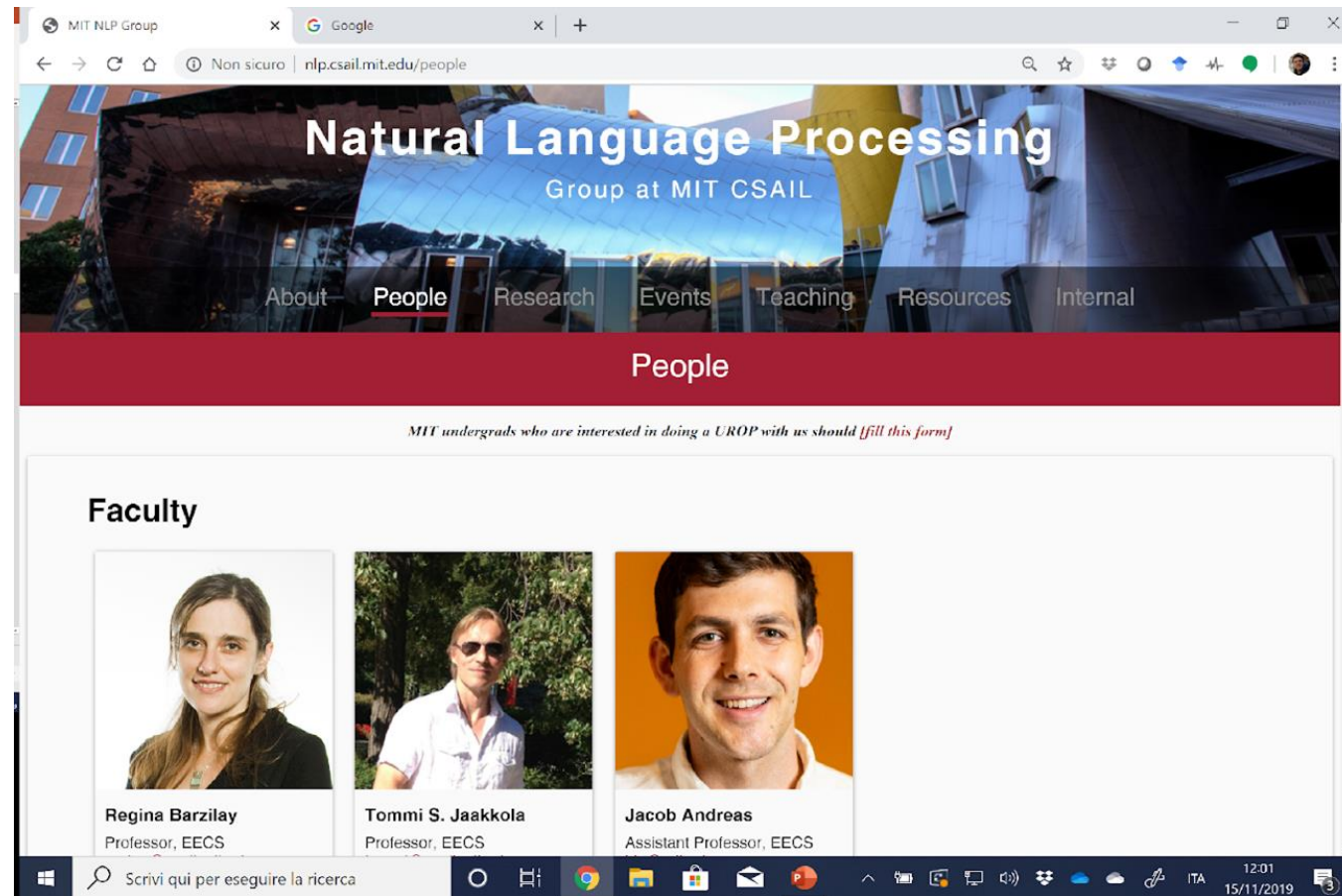
Articolo

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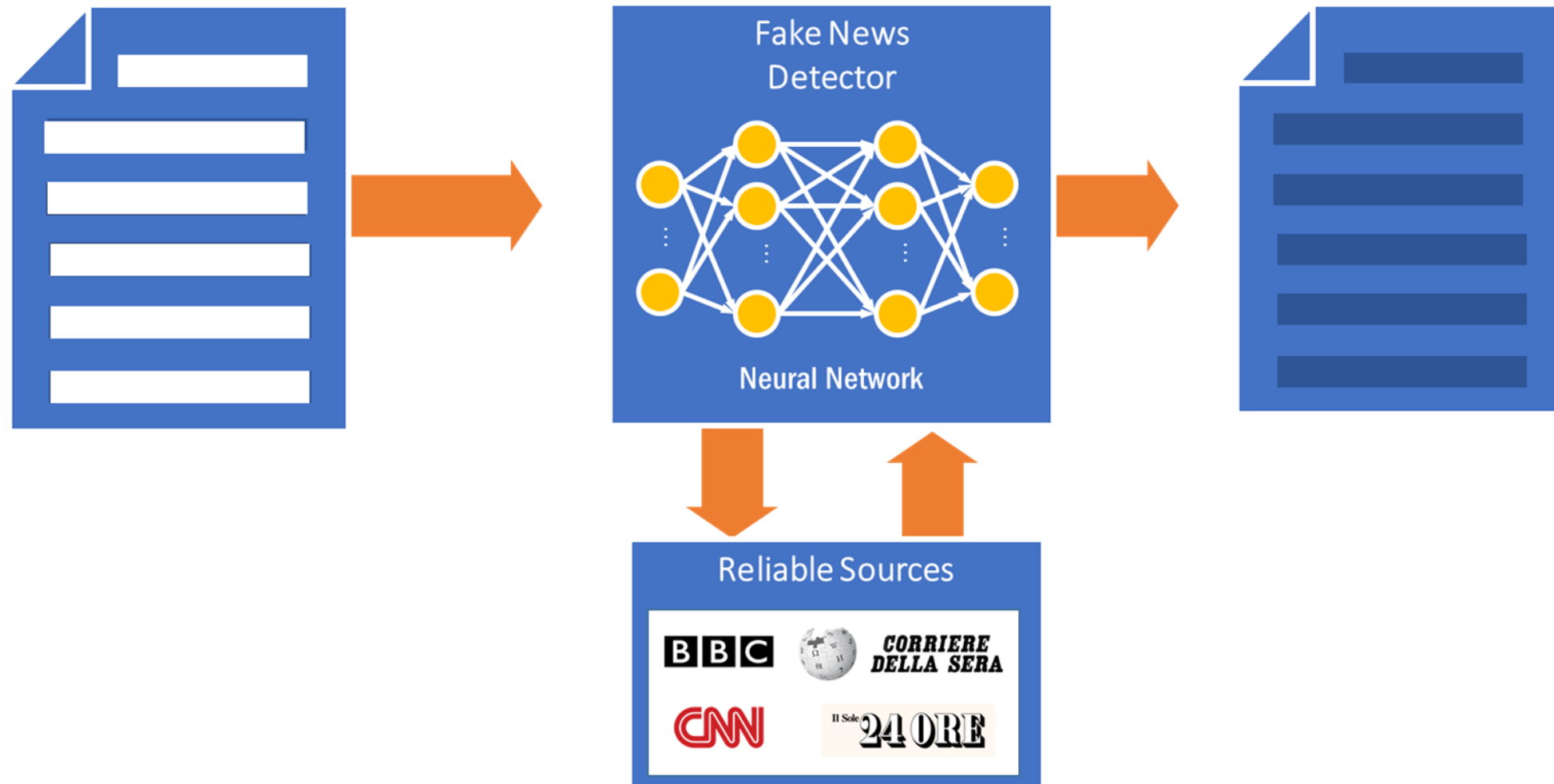


Fake News Project

Project in collaboration with Massachusetts Institute of Technology (MIT)



Fake News Project



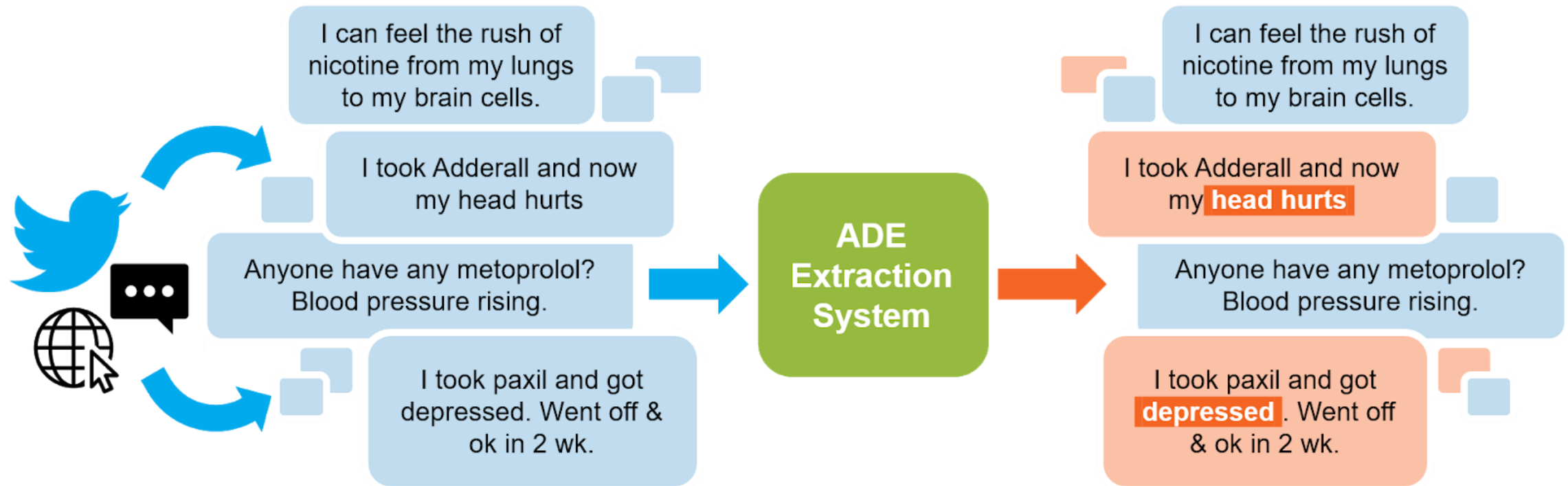
COVID-19 Vaccine

This project aims at monitoring and visualizing the dynamics of the Twitter debate, to get a sense of the public sentiment, while highlighting tweets that mention potential adverse events.



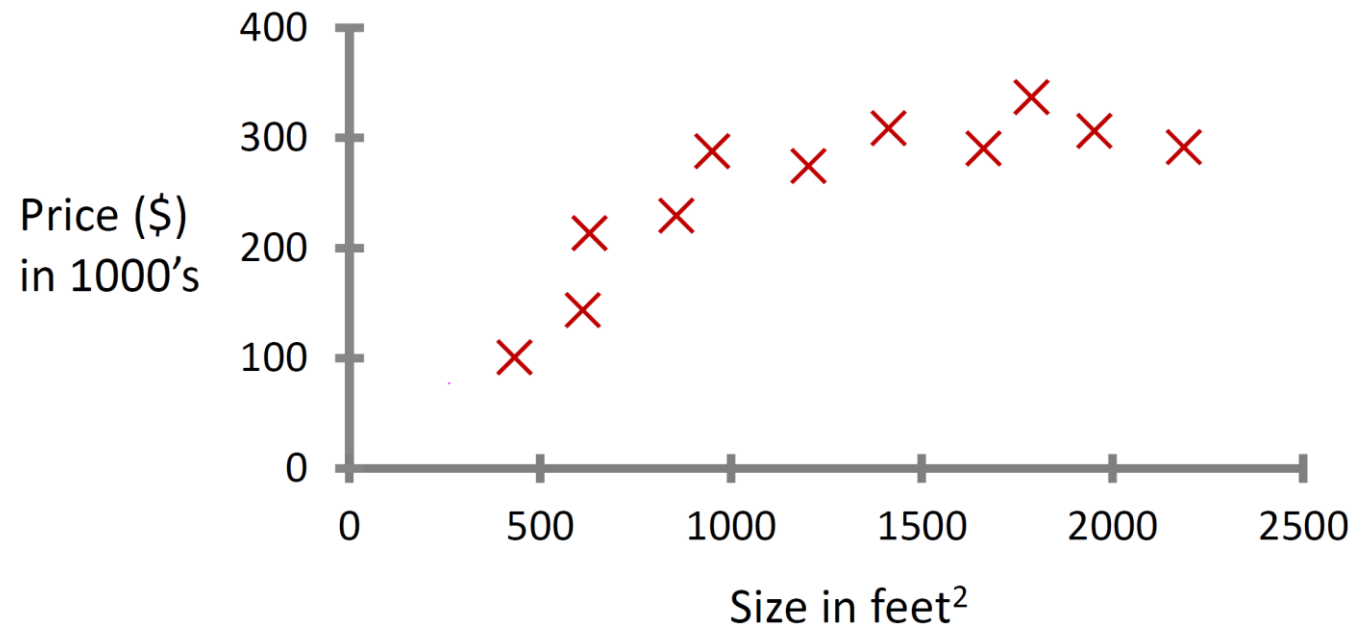
<http://ailab.uniud.it/covid-vaccines/>

COVID-19 Vaccine



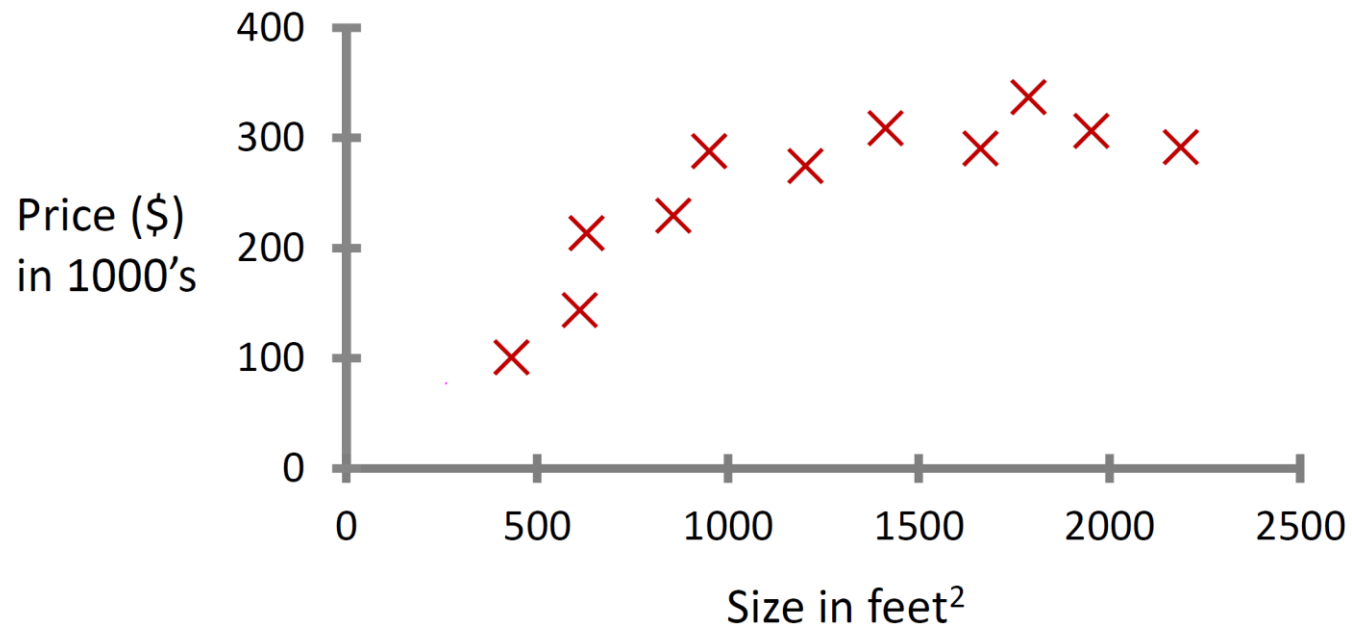
Regression

Housing price prediction



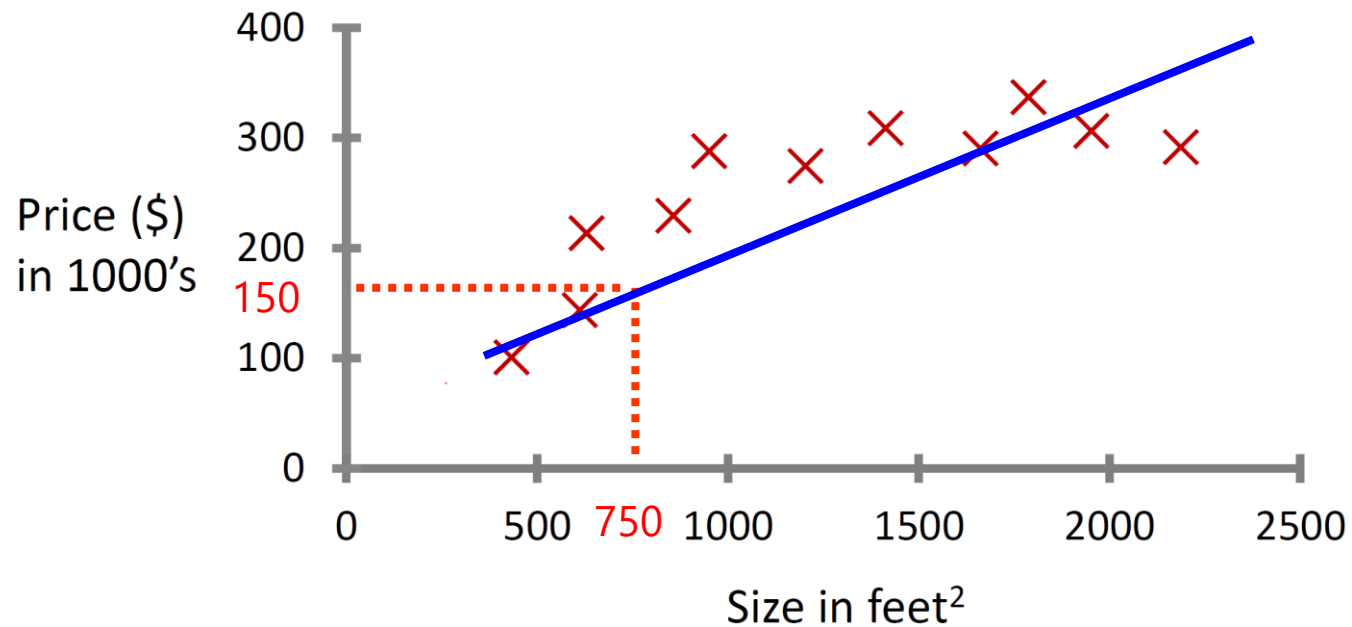
Housing price prediction

Suppose you have a friend with a house of 750 feet². How can a learning algorithm help you?



Housing price prediction

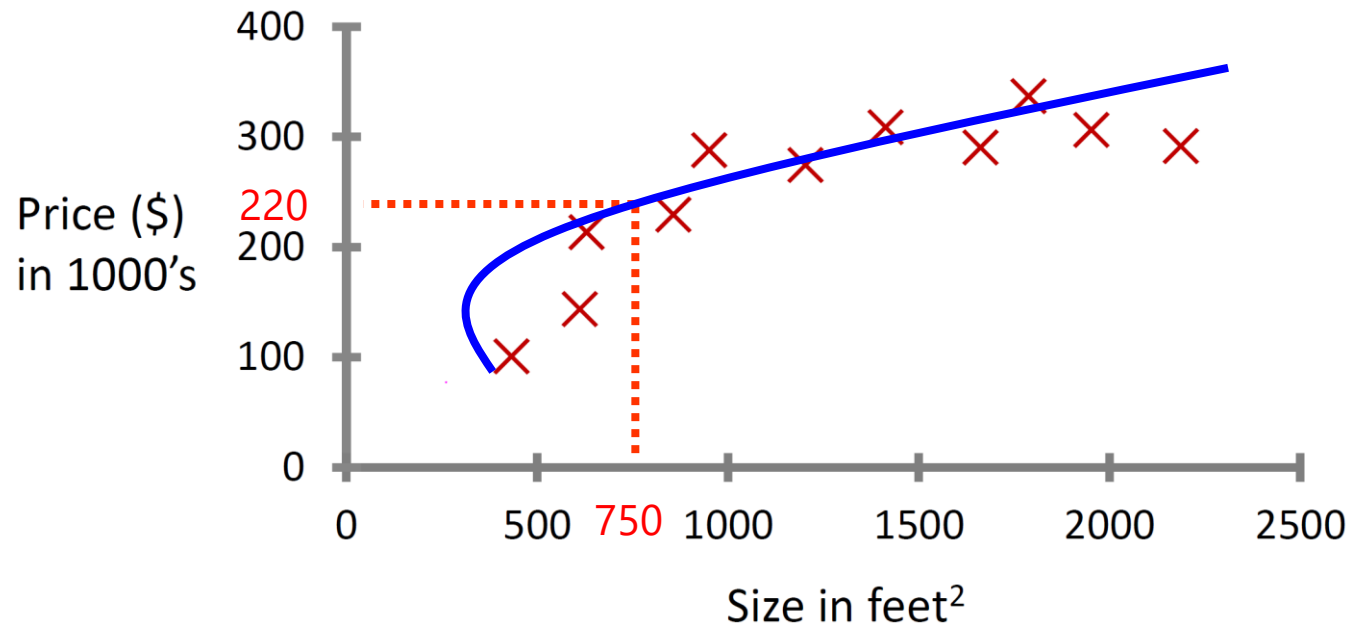
Suppose you have a friend with a house of 750 feet². How can a learning algorithm help you?



A learning algorithm can fit a straight line to the data. Based on that, it looks like maybe the house can be sold for about \$ 150,000.

Housing price prediction

Suppose you have a friend with a house of 750 feet². How can a learning algorithm help you?



A learning algorithm can fit a quadratic function or a second-order polynomial to the data. Based on that, it looks like maybe the house can be sold for about \$ 220,000.

Supervised Learning

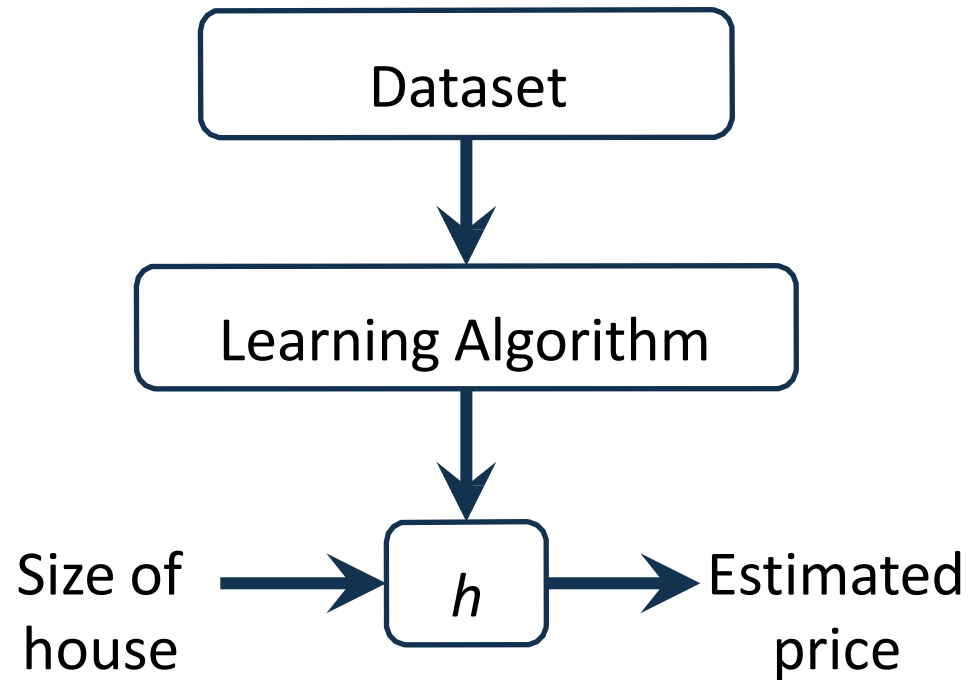
The supervised learning refers to the fact that we use a dataset in which "the right answer" are given.

Dataset

Size in feet ² (x)	Price (\$) in 1000's (y)
2104	460
1416	232
1534	315
852	178
...	...

This algorithm is called regression algorithm, which given an input value (size in feet) it predicts a continuous value as output (price)

Supervised Learning

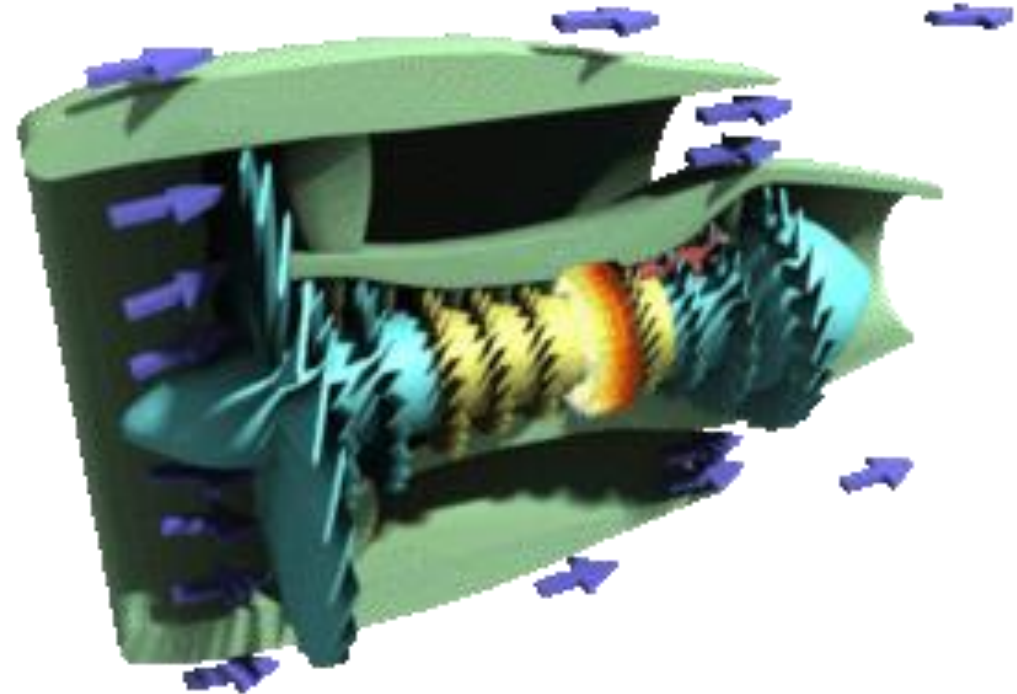


h = hypothesis

Examples



Predictive Maintenance



Test

You're running a company, and you want to develop learning algorithms to address each of two problems.

Problem 1: You have a large inventory of identical items. You want to predict how many of these items will sell over the next 3 months.

Problem 2: You'd like software to examine individual customer accounts, and for each account decide if it has been hacked/compromised.

Should you treat these as classification or as regression problems?

- ☐ Treat both as classification problems
- ☐ Treat problem 1 as a classification problem, problem 2 as a regression problem
- ☒ **Treat problem 1 as a regression problem, problem 2 as a classification problem**
- ☐ Treat both as regression problems.

Observations

Suppose you have a collection of swan



A supervised learning algorithm will “learn” that swan is white

Observations

However, inductive conclusions can not be accepted as certain!



Future observations could be in conflict with the generalization, and a revision would be required.

Training and Test Sets

The set of available examples is usually partitioned into:

- TRAINING SET, constituted by a part of the available examples and exploited to learn the hypothesis
- TEST SET, constituted by the remaining part of the available examples and exploited for evaluating the accuracy of the learned hypothesis

Note the success depends directly on the partitioning criteria (into training and test set)

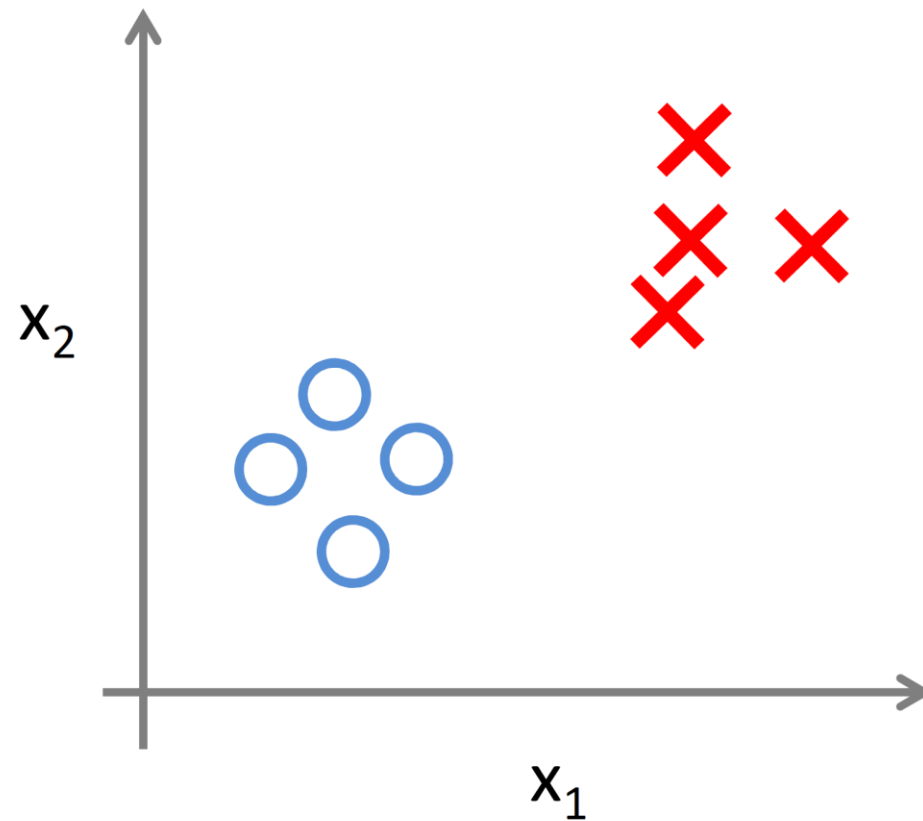
The training set should well balanced over all the possible cases of the phenomenon, otherwise there is the risk of a partial learning (as seen before).

Such situation is called OVERFITTING

Unsupervised Learning

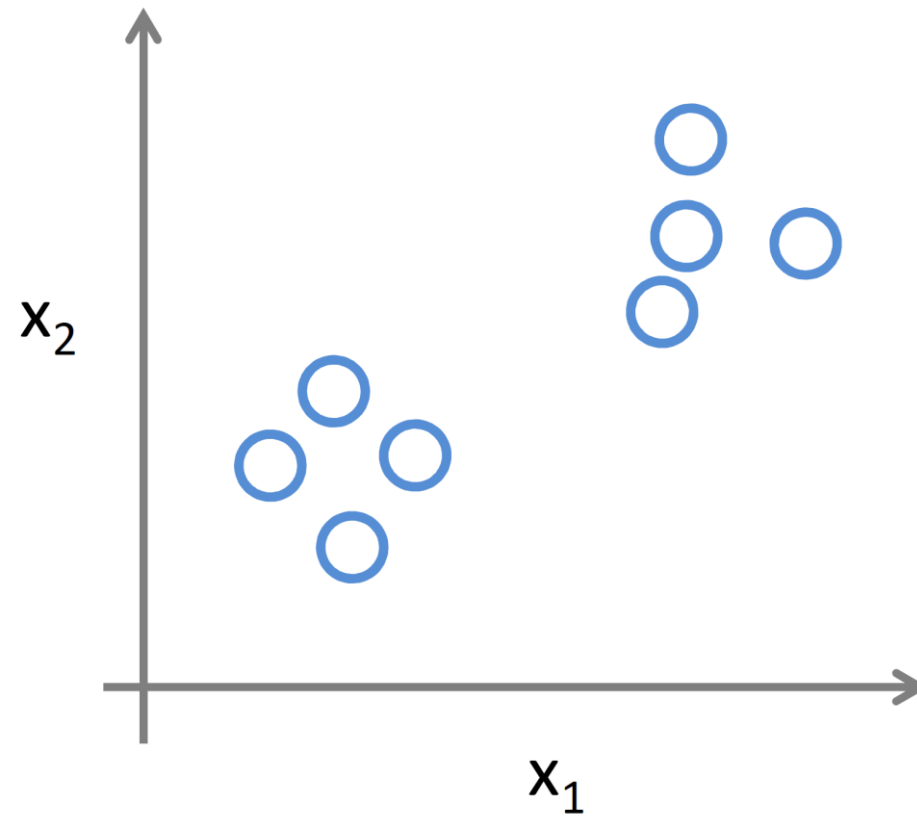
Supervised Learning

Recall, in supervised problem each example is labelled either as a positive or as negative



Unsupervised Learning

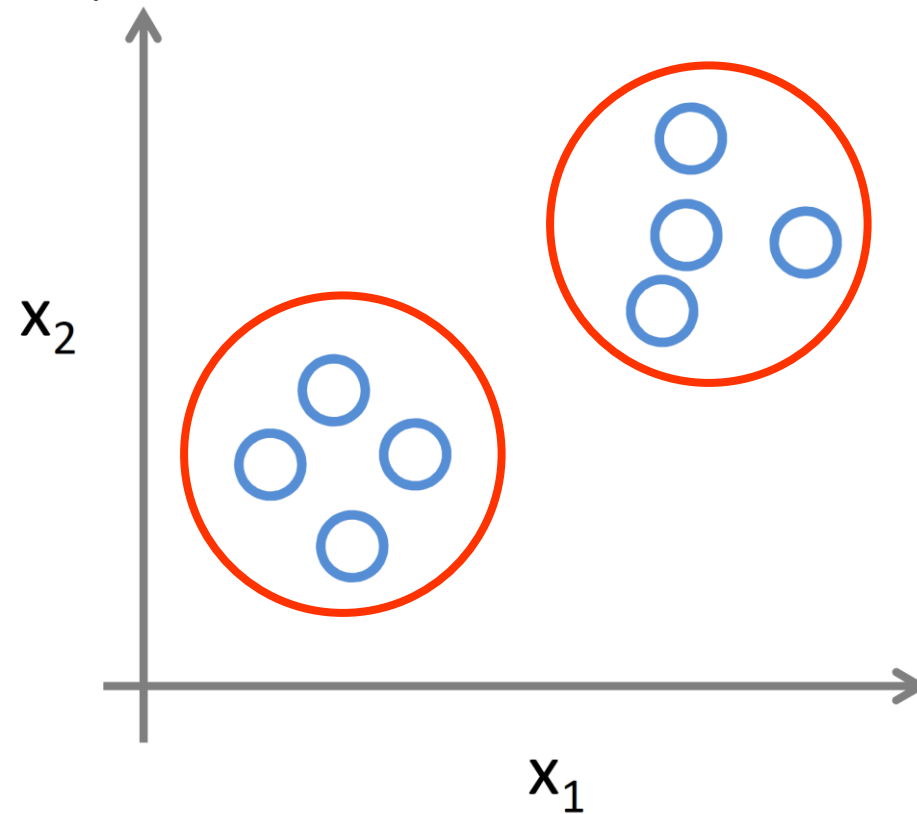
In unsupervised problem labels do not have any labels.



Unsupervised Learning

How can we find a structure of these data?

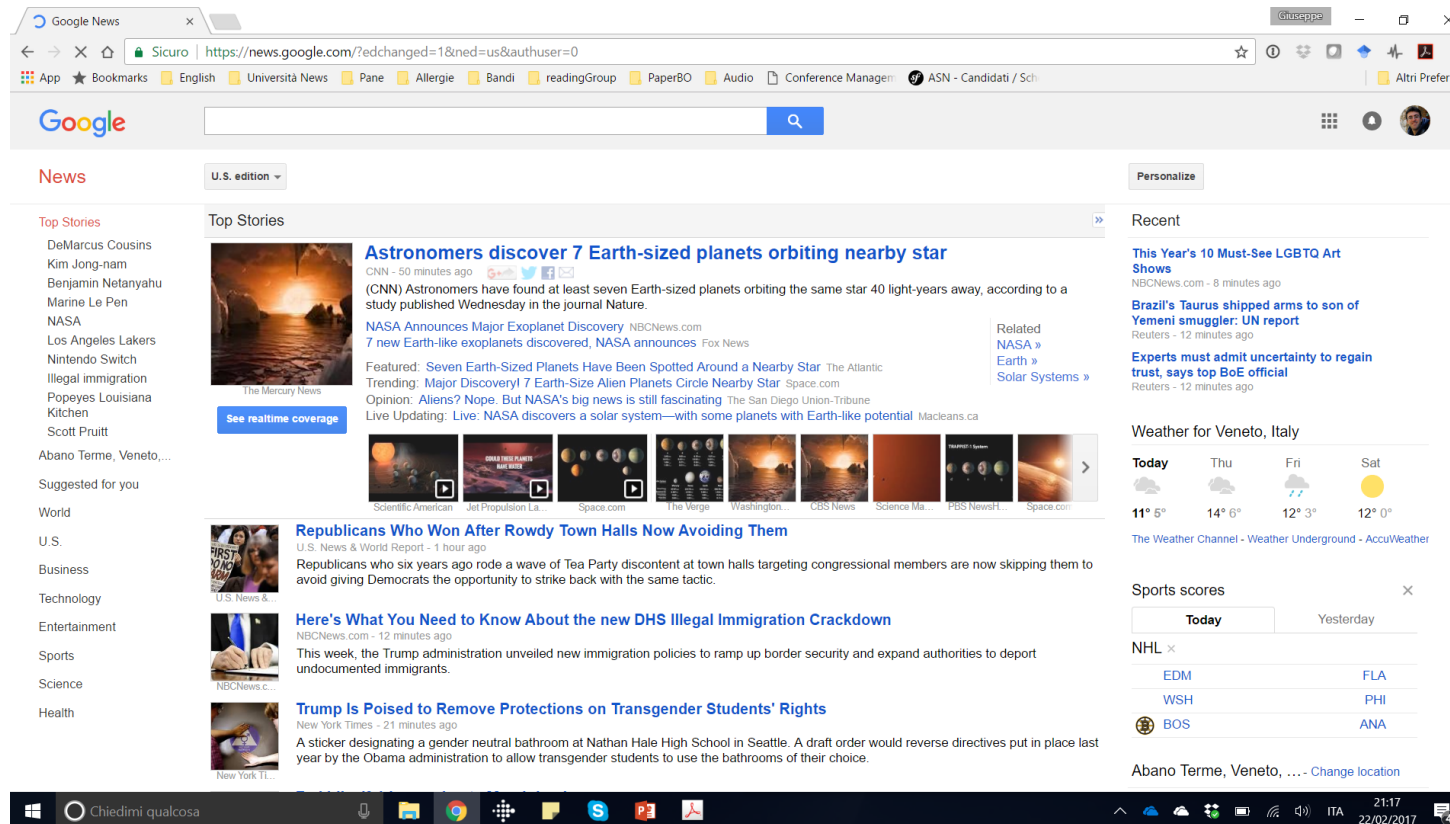
An unsupervised learning algorithm might decide that these data live in two different clusters (clustering algorithm)



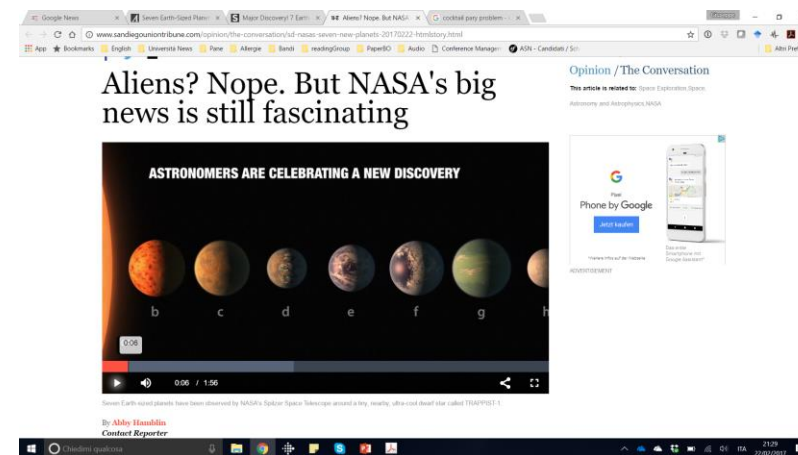
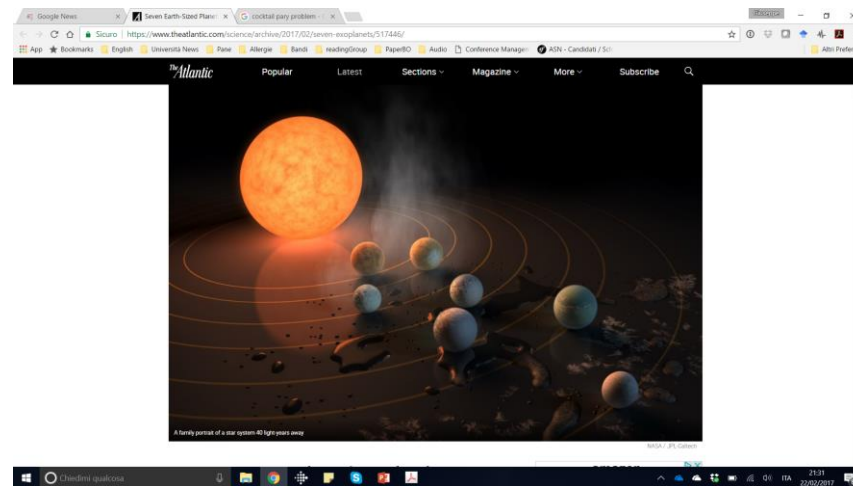
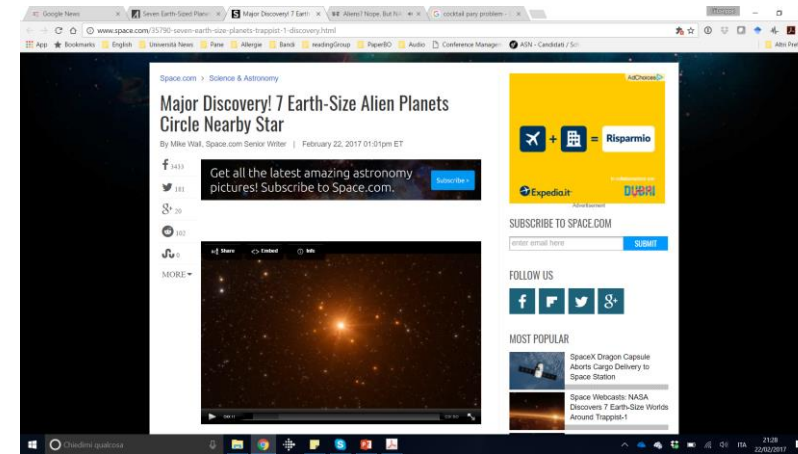
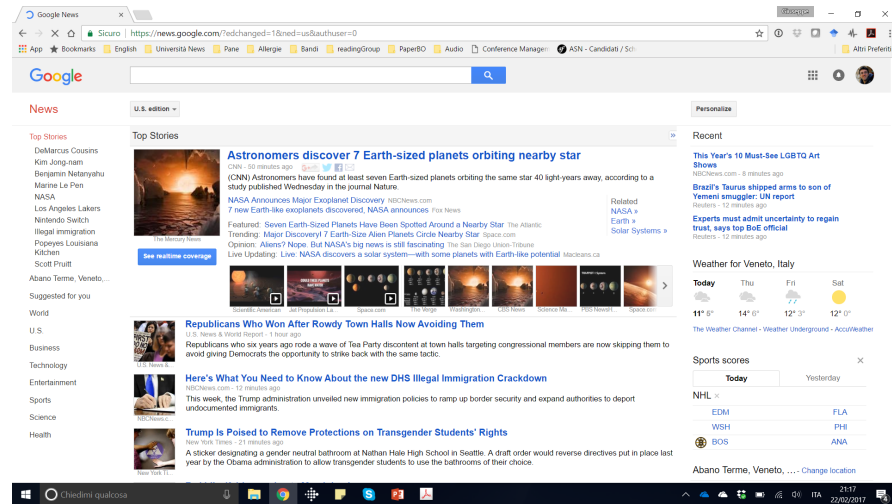
Examples

Cluster algorithms can be used in different applications

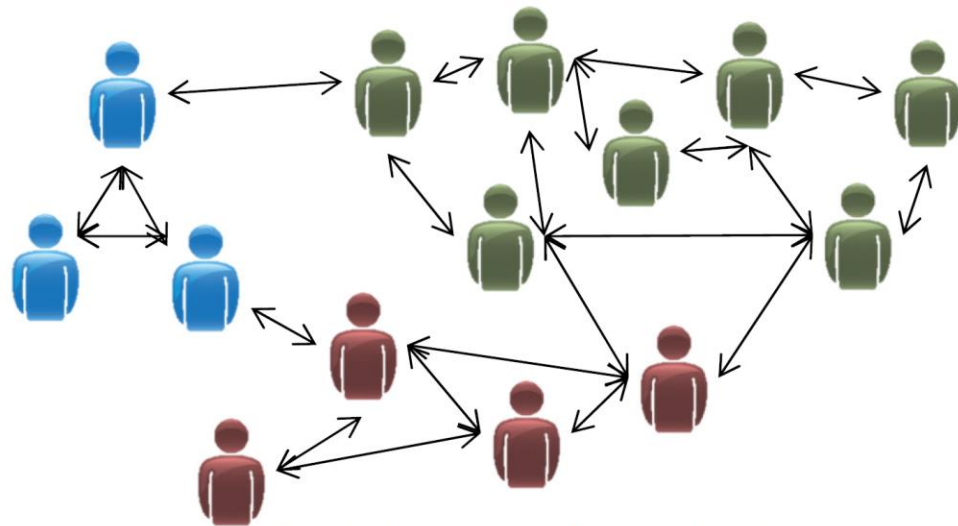
Everyday Google News analyses thousands of thousands of new stories and groups them into cohesive stories.



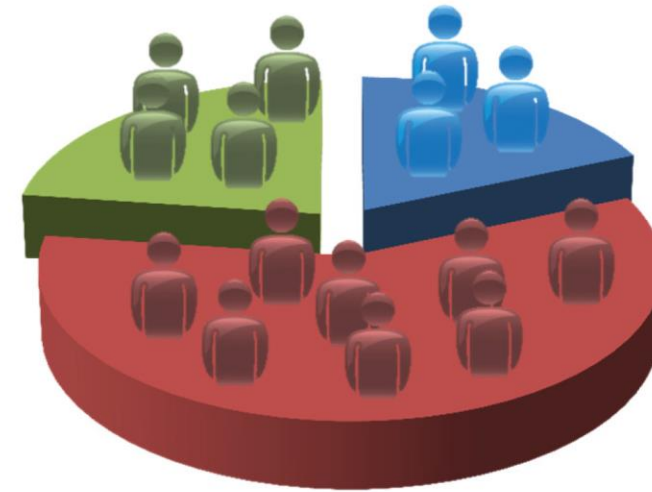
Examples



Examples



Social network analysis



Market segmentation

Of the following examples, which would you address using an unsupervised learning algorithm? (Check all that apply.)

- ☐ Given email labeled as spam/not spam, learn a spam filter
- ☐ Given a set of news articles found on the web, group them into set of articles about the same story.
- ☐ Given a database of customer data, automatically discover market segments and group customers into different market segments.
- ☐ Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not.

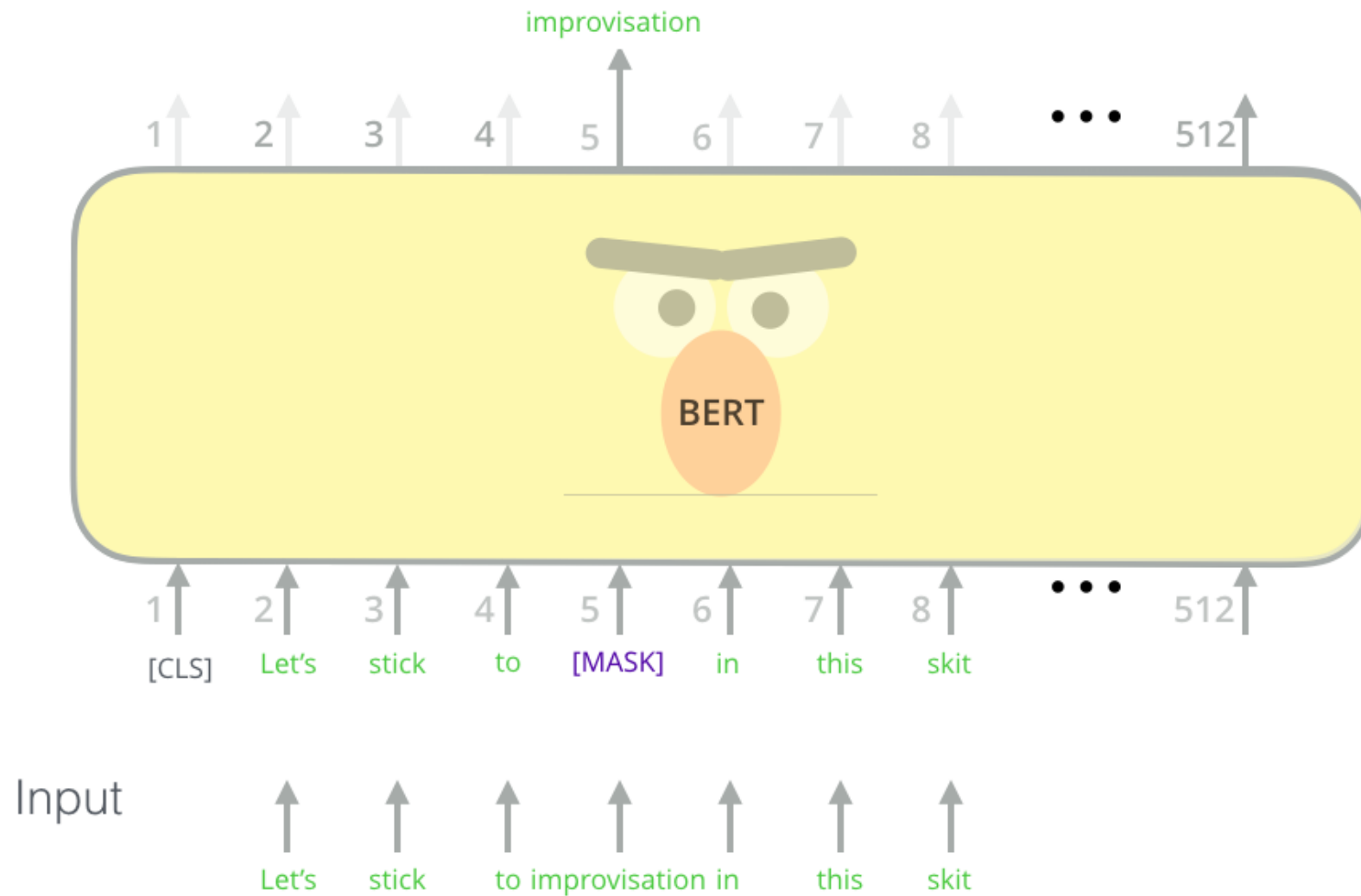
Test

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- ☐ Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not.

Self Supervised Learning

BERT – Language Model



Reinforcement Learning

Basic Concept



Learning Hand-Eye Coordination for Robotic Grasping with Deep Learning and Large-Scale Data Collection

Sergey Levine Peter Pastor
Alex Krizhevsky Deirdre Quillen
Google

The Game of Go

In 2016 AlphaGo (Deep Mind) won Lee Sedol, the world's top Go player.



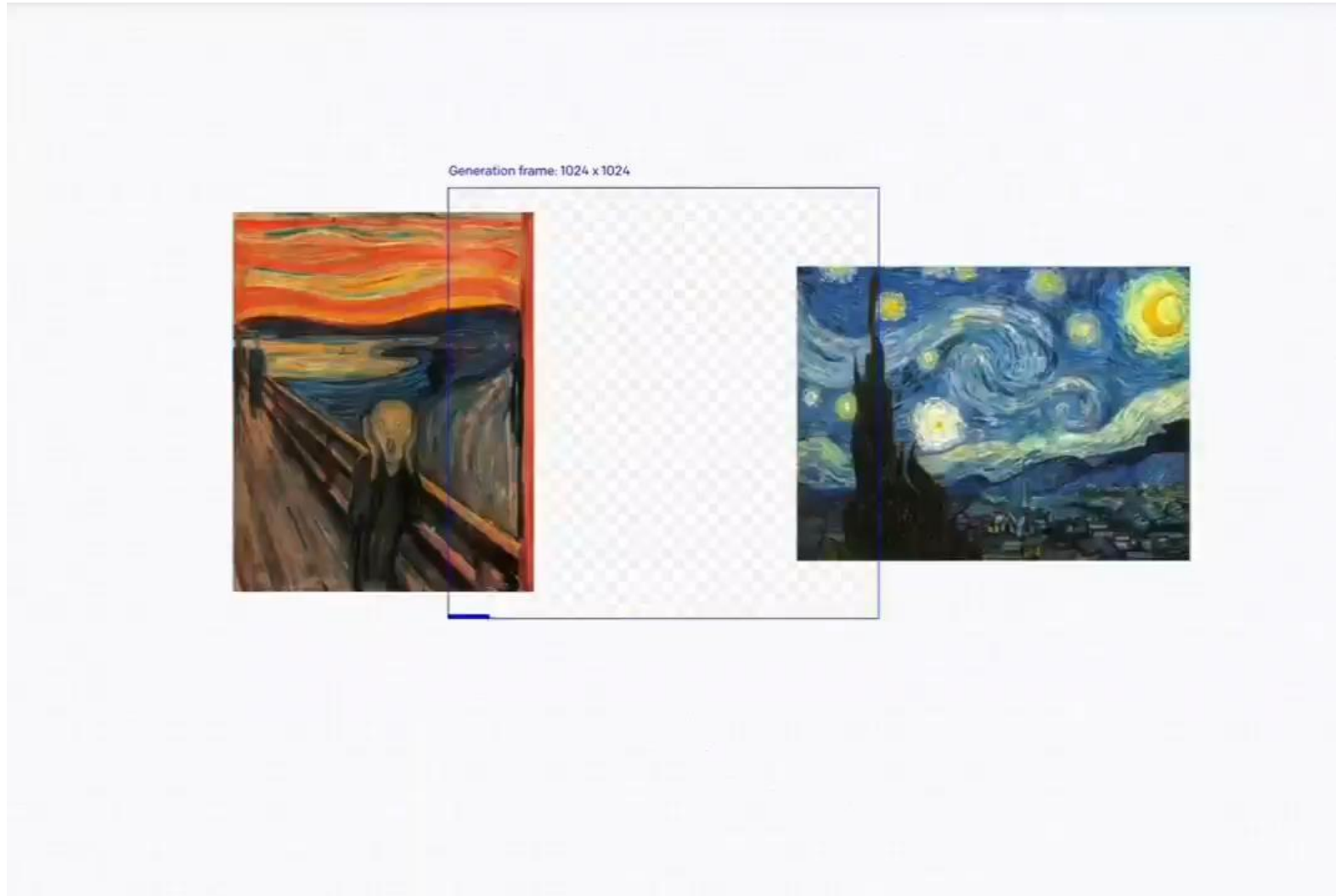
See more info:

<http://airesearch.com/tag/go/>

[AlphaGo – The Movie \(Full Documentary\)](#)

Content Generation

Generative Adversarial Networks (GANs)



Content Generation

